MID-TERM/Minor EXAMINATION
B.TECH (March, 2022)

Subject Code: BCS 204	**Subject: Design and Analysis of algorithm**
Time : 1 ½ Hours	**Maximum Marks : 30**
Note:Q. 1 is compulsory. Attempt any two questions from the rest.	

Q1		**(5*2)**
	(a) Let Q be a QuickSort Program to sort numbers in ascending order using the first element as pivot. Calculate the number of comparisons t1 and t2 made by Q for the inputs {11, 22, 33, 44, 55} and {24, 21,25, 23, 22} respectively. Which one is greater, t1 or t2 ?	
	(b) Solve the following recurrence relation $T(n) = T(n/5) + T(4n/5) + n$	
Q2		**(5,4,1)**
	(a)Write recursive and non-recursive algorithms for generating the Fibonacci series. (b)Analysis the time complexity of both the algorithms (c) Which one is efficient?	
Q3		**(7,3)**
	Let T[1..n] be the sorted array of distinct integers, some of which may be negative. (a) Give an algorithm that can find an index i such that $1 \leq i \leq n$ and T[i]=i, provided such an index exists. Your algorithm should take a time in O(log n) in the worst case. (b) Analyse the complexity of your algorithm and show that it is O(log n)	
Q4		**(7,3)**
	Let P1, P2…. Pn be n programs to be stored on a disk. Program Pi requires Si kilobytes of storage and the capacity of the disk is D kilobytes where $D < \sum_{i-1}^{n} Si$. We want to maximise the number of programs held on the disk. (a)Write a greedy algorithm and (b) Analyze its complexity.	